

TradeAux Firebase Backend — Deployment Guide

File Structure

```
your-project/
├─ firebase.json
├─ .firebaserc
├─ firestore.indexes.json    ← Composite indexes
├─ firestore.rules          ← Security rules
├─ functions/
│   ├─ index.js             ← Cloud Functions
│   └─ package.json         ← firebase-functions + firebase-admin
├─ set-admin.js             ← One-time admin setup script
└─ public/
    ├─ signin.html
    ├─ seller-auction.html   ← The completed auction floor
    └─ ...
```

Prerequisites

1. Install Firebase CLI: `npm install -g firebase-tools`
2. Login: `firebase login`
3. Initialize: `firebase init` (select Firestore, Functions, Hosting)
4. Download service account key from Firebase Console → Project Settings → Service Accounts → Generate New Private Key. Save as `serviceAccountKey.json` in project root (for `set-admin.js` only — do NOT commit to git).

Deployment Commands (run in order)

```
# 1. Deploy Firestore indexes (required for bid sorting queries)
firebase deploy --only firestore:indexes

# 2. Deploy security rules
firebase deploy --only firestore:rules

# 3. Deploy Cloud Functions
firebase deploy --only functions

# 4. Deploy hosting (HTML files)
firebase deploy --only hosting
```

One-Time Admin Setup

After deploying functions, run this locally to give yourself admin access:

```
# Replace YOUR_UID with your actual Firebase Auth UID
node set-admin.js YOUR_UID
```

Find your UID in Firebase Console → Authentication → Users.

Cloud Functions Summary

Function	Type	Purpose
acceptBid	HTTPS Callable	Seller accepts a bid, deducts inventory, captures \$25 fee
runDuhe	Pub/Sub Scheduled	Auto-ends expired auctions every 15 min, declines orphaned bids
runDuheManual	HTTPS Callable	Admin manually triggers cleanup
getSellerStats	HTTPS Callable	Returns seller dashboard metrics

Firestore Collections

Collection	Purpose
profiles	User profiles (sellers & buyers)
seller_briefs	Auction listings with quantity, status, fees
bids	Buyer offers with price, quantity, status
fee_records	\$25 fee transactions (auto-created on accept)

Security Rules Highlights

- Sellers fully own their briefs (CRUD)
- Buyers can read `active` and `auctioning` briefs
- Bids are read-only for everyone; writes only via `acceptBid` Cloud Function
- Fee records are write-locked to Cloud Functions